Computer Vision

To extract information from images and videos

Digital computer having human like vision

Applications

- Robotics
- Autonomous Vehicle
- Biomedical
- Internet media

Signal

Function of Independent Variables

1D - Audio - $y = f(t)$ t: time

2D - Image - $y = f(u, v)$ u, v: Spatial Coordinates

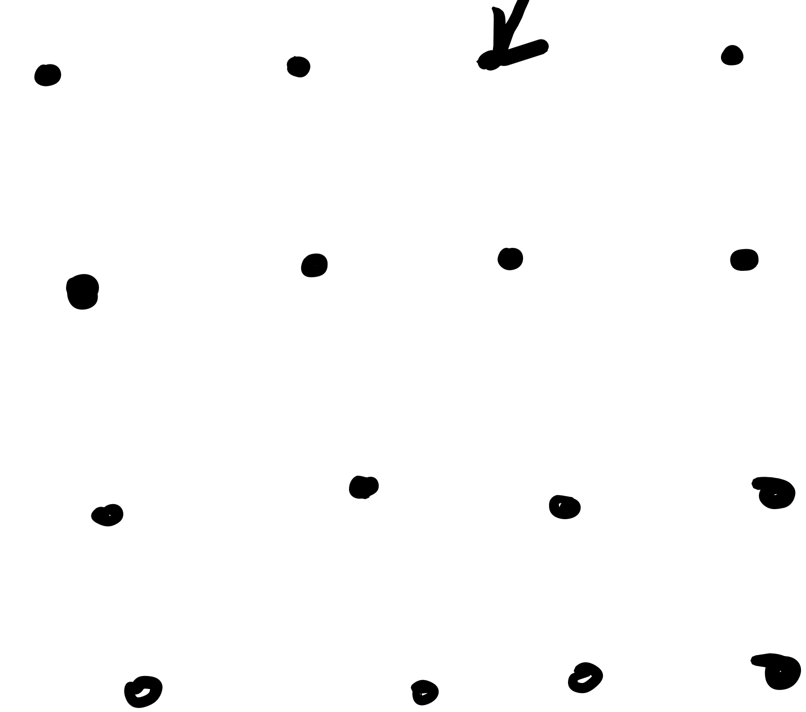
3D - Video - $y = f(u, v, t)$

Image as Signal

set of intensity levels at spatial grid points

$I(x, y)$

grid ⇒ (x, y)



4x4 Image

Matrix Description

Image is usually stored as 2D array

eg (in python)

[[1, 2, 10, 11], [12, 25, 46, 37],
[32, 43, 47, 48],
[27, 22, 23, 0]]

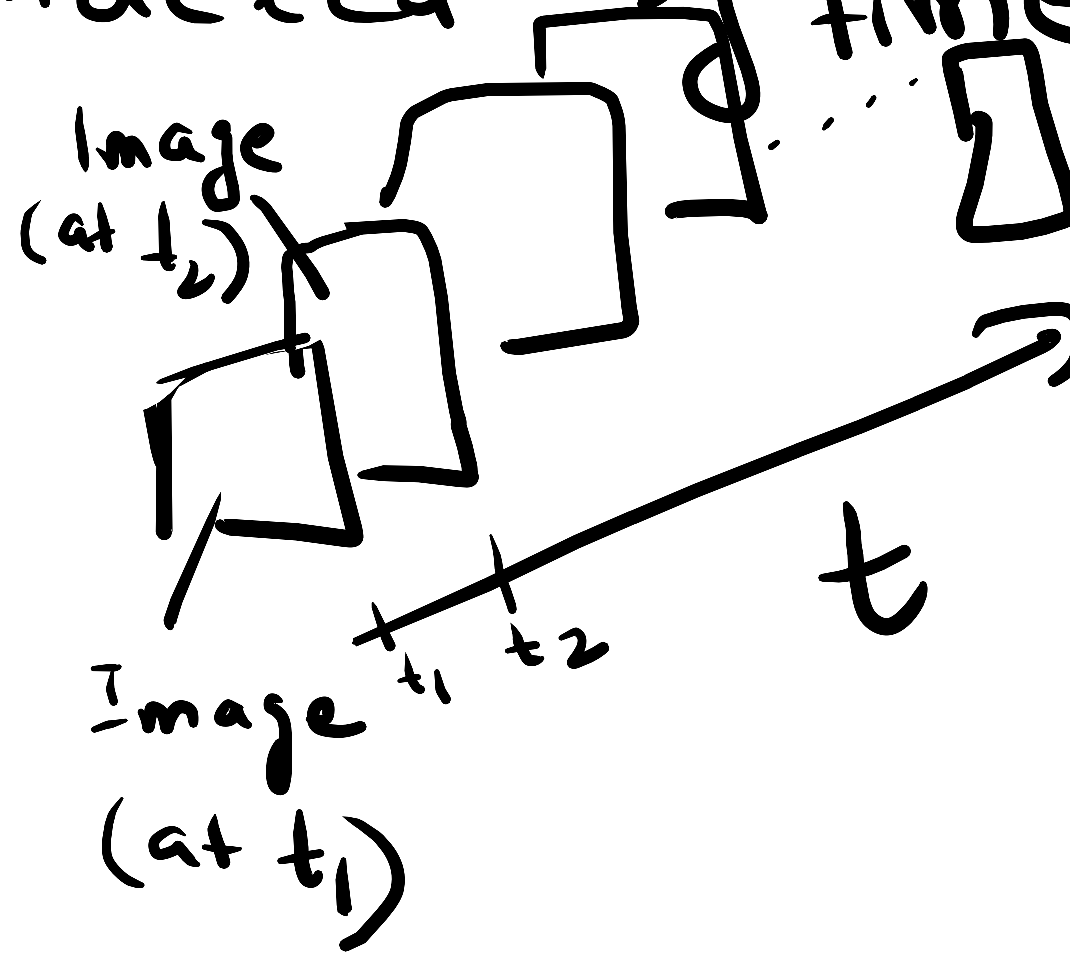
Representation of 4x4 image as list of lists in Python

Video as Signal

Sequence of Images indexed by time

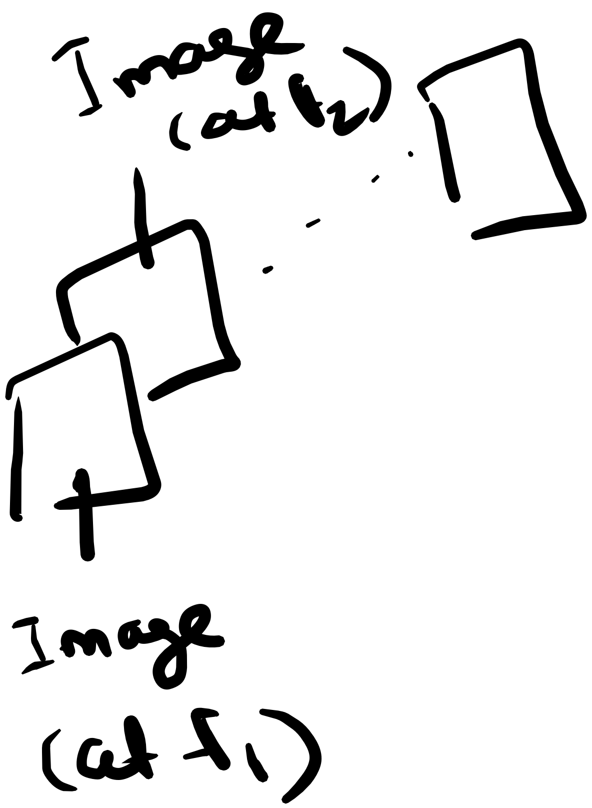
In programming

Temporally related Matrices



nD Signals

Satellite Images Captured at various frequencies are an example for nD signals



$$y = f(\underbrace{x, y}_{\text{spatial coordinates}}, \underbrace{f_1, f_2, \ldots, f_n}_{n \text{ frequencies}})$$

Image formats

- jpg
- gif
- bmp (bitmap image)
- png
- tiff

Video formats

- .mp4
- .mkv
- .mov (Apple Quicktime)
- .webm (by Google for online Streaming)

Types of Images

- Binary Image
  - Intensity levels
  - $[0, 1]$
- Grayscale Image
  - Intensity levels
  - $[0, 1, 2, \dots 255]$
- Color Image
  - Three channels
  - R G B
  - $[I_1]$ $[I_2]$ $[I_3]$
  - $I_1 \in [0, 1, 2, \dots 255]$
  - $I_2 \in [0, 1, 2, \dots, 255]$
  - $I_3 \in [0, 1, 2, \dots 255]$

Lab: Matrix Operation

Common steps in CV Algo

Read image or video and store in Matrix

```
img = iio.imread(img.tiff)
img2 = np.copy(img)
n = np.shape(img)
```

Extract metadata

```
nrow = n[0]
ncol = n[1]
```

Processing on specific pixels using iterative loops

```
for i in range(1, nrow-1):
    for j in range(1, ncol-1):
        img2[i,j] =
```

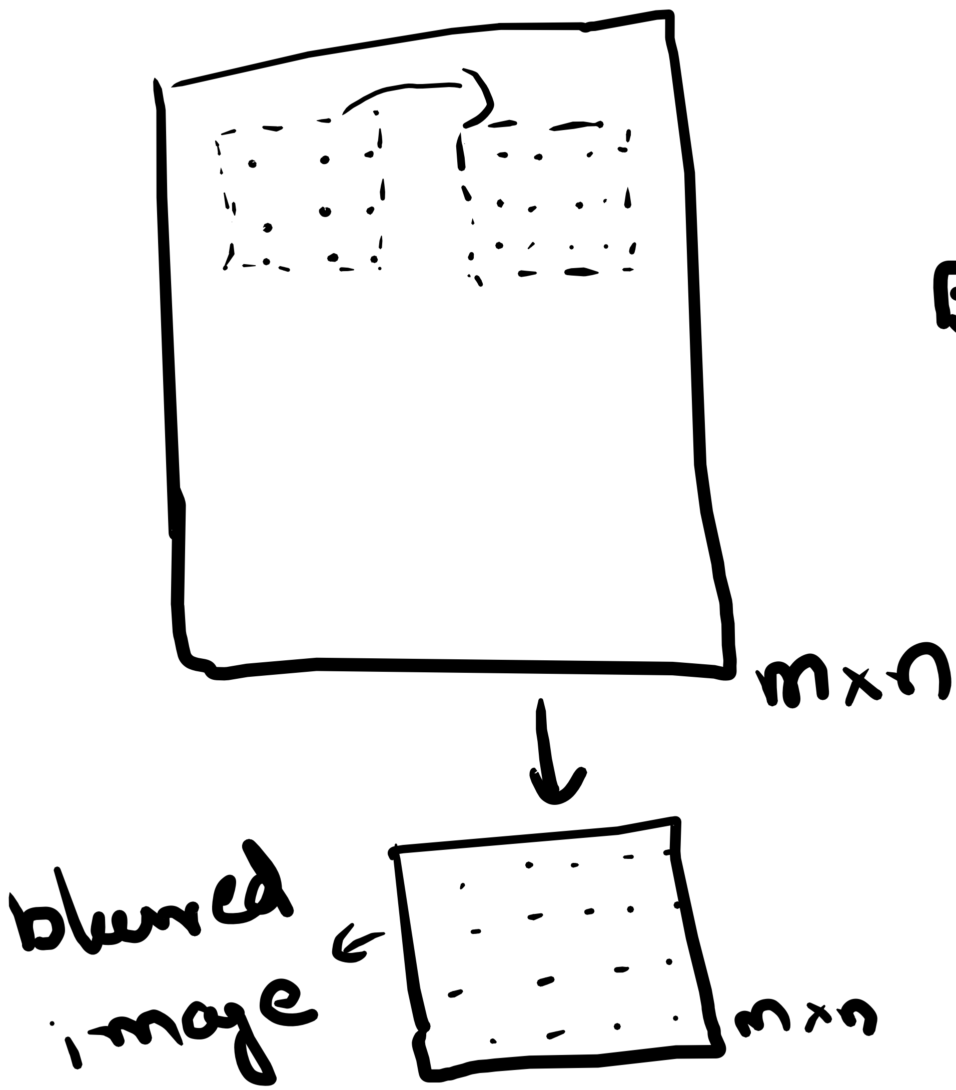
← Processing Step

plot output

$$h = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Spatial Filtering

Identify a Mask – 3x3 or 5x5 or 7x7



Blurring – Mean of pixel elements inside mask

The mask is slided over the entire image to generate a blurred image.